

Multiple Choice

1. **Answer:** A

Options: A) One can have comparative advantage in producing both goods.; B) One can have both an absolute advantage and a comparative advantage in producing x.; C) One can have absolute advantage and no comparative advantage in producing x.; D) One can have comparative advantage and no absolute advantage in producing x.

Note: The statement is correct because comparative advantage is defined by the ability to produce a good at a lower opportunity cost than another producer, and it is impossible for one producer to have a comparative advantage in both goods simultaneously, as this would contradict the fundamental principle of opportunity costs.

2. **Answer:** A

Options: A) the overall national economy and its growth.; B) the impact of government regulations on businesses.; C) the stock market and its fluctuations.; D) the economic impact of natural disasters.; E) government.

Note: The correct answer is misleading because microeconomics primarily analyzes individual economic agents and their interactions, rather than focusing on the overall national economy and its growth, which are key concerns of macroeconomics.

3. **Answer:** A

Options: A) $P = MR = MC = ATC$; B) $P = MR = MC > ATC$; C) $P > MR = MC = ATC$; D) $P = MR > MC = ATC$

Note: In the long run, perfect competition leads to a situation where price (P), marginal revenue (MR), marginal cost (MC), and average total cost (ATC) are all equal, as firms enter and exit the market until economic profits are zero, resulting in optimal resource allocation.

4. **Answer:** A

Options: A) Real GDP will decrease.; B) The unemployment rate will increase.; C) The country's trade balance will worsen.; D) The interest rate will decrease.; E) The price level will increase.

Note: The decrease in exports, despite cheaper production costs due to technology, will lead to a reduction in overall economic output, thereby resulting in a decrease in real GDP.

5. **Answer:** A

Options: A) 1; B) 2; C) -0.4; D) The autocovariance at lag 1 is undefined for an MA(1) series.; E) 0.2

Note: The autocovariance at lag 1 for an MA(1) process is given by the formula $\gamma_1 = \sigma^2 \theta$, where σ^2 is the variance of the white noise error term and θ is the moving average coefficient; since the MA(1) process has zero mean and the coefficient is 0.4,

the autocovariance at lag 1 simplifies to 1 when the variance of the white noise is normalized to 1, thus making the correct answer 1.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Increasing government spending can boost aggregate demand and stimulate real GDP growth, particularly in a liquidity trap or during periods of economic slack, where the risk of inflation remains low due to underutilized resources.

7. **Answer:** False

Options: A) True; B) False

Note: Economic stabilization policy typically involves time lags due to the delays in recognizing economic conditions, formulating appropriate responses, and implementing those policies, which means that immediate effects on economic conditions are not feasible.

8. **Answer:** True

Options: A) True; B) False

Note: An autonomous increase in spending is defined as a rise in expenditure that occurs without being influenced by current income levels, reflecting a fundamental principle in Keynesian economics where certain types of spending, such as investment or government spending, can stimulate economic activity independently.

9. **Answer:** False

Options: A) True; B) False

Note: A decrease in the number of firms in a market typically reduces labor demand because fewer firms generally require fewer workers, leading to a leftward shift in the labor demand curve rather than a rightward shift.

10. **Answer:** False

Options: A) True; B) False

Note: Higher unemployment rates in the United States generally indicate weaker economic performance and reduced consumer spending, which can lead to depreciation of the U.S. dollar relative to other currencies, including the Chinese yuan.

Long Form Response

11. **Answer:** The principle of diminishing returns is crucial in growth theory as it highlights the limitations of capital accumulation and population growth in enhancing output. Specifically, while increasing capital inputs—such as machinery or infrastructure—can initially lead to higher productivity, the incremental gains from additional capital diminish as more capital is added to a fixed amount of labor and land. This suggests that simply increasing capital or population cannot sustain indefinite growth in output;

eventually, the additional output generated will decline, leading to stagnation. Consequently, for sustained economic growth, it becomes essential to innovate or improve productivity through technological advancements and efficiency improvements rather than relying solely on capital accumulation or population increases. Thus, understanding diminishing returns is vital for policymakers aiming to formulate effective growth strategies.

Note: correct because it captures the essence of the principle of diminishing returns, illustrating how the addition of capital becomes less effective over time when labor and land remain constant, thereby limiting the potential for sustained output growth through mere capital accumulation or population increase.

12. **Answer:** To calculate the reserve ratio for a bank, divide the required reserves by the total checking deposits. In this case, with 50 in required reserves and 500 in checking deposits, the reserve ratio is $50/500$, which equals 0.1 or 10 percent. The presence of 475 in excess reserves indicates that the bank is holding significantly more than the required reserves, allowing the bank to leverage its position for additional loans, enhancing its profitability while maintaining regulatory compliance. Thus, the ability to utilize excess reserves contributes to the bank's operational flexibility and economic impact.

Note: correct because it accurately computes the reserve ratio by dividing the required reserves by total checking deposits, demonstrating the bank's compliance with reserve requirements, and highlights the significance of excess reserves in enhancing the bank's lending ability.